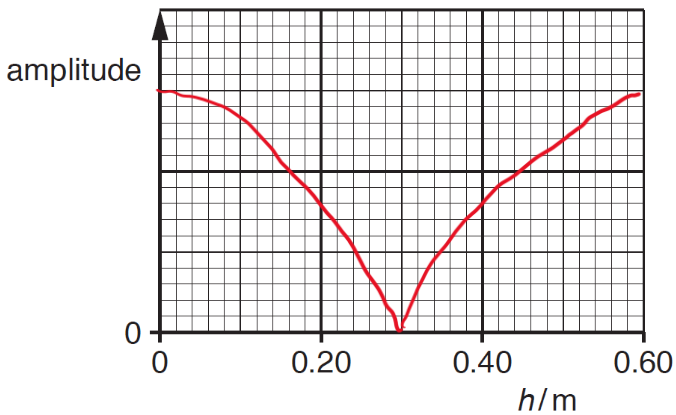
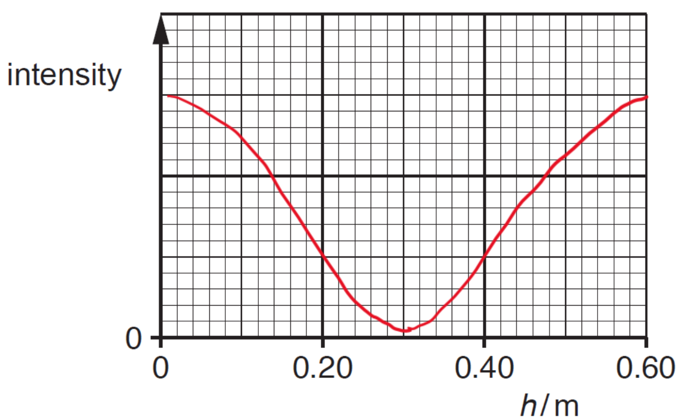
Qns	Answer	Marks
1(a)	(incident) wave reflects at end/top of tube (incident) wave and reflected wave superpose waves have same frequency, wavelength and speed	B1 B1 B1
1(b)(i)	line has maximum value of amplitude at h = 0 and h = 0.60 m only AND line has minimum/zero value of amplitude at h = 0.30 m only modulus cosine graph (Sharp at h = 0.30 m) 	M1 A1
1(b)(ii)	line has maximum value of amplitude at h = 0 and h = 0.60 m only AND line has minimum/zero value of amplitude at h = 0.30 m only AND modulus cosine-squared graph 	B1
1(c)(i)	vertical/along length of tube/along axis of tube	B1
1(c)(ii)	phase difference = 0	A1
1(c)(iii)	phase difference = 180°	A1
1(d)	$v = f\lambda$ $f = 340 / (2 \times 0.60)$	C1

Qns	Answer	Marks
	= 280 Hz	**A1**
1(e)(i)	*f* = 340 / 0.60 = 570 Hz (or 2 x 280 = 560 Hz)	 **A1**
1(e)(ii)	0.60 / 4 = 0.15 m	**A1**